VICTORY 2027

Conventional Sources Of Energy

LECTURE-2 Geography

Sameer Sir



Topics To Be Covered

1. Natural Gas
2. Hydel Power
3. General service enlistment Act.
   - ↳ Indian Sepoys were forced to go on Sea Voyages.
4. Brahmo Samaj

Any two { Moderate / Early nationalist
Extremist /

Mineral and Energy Resources

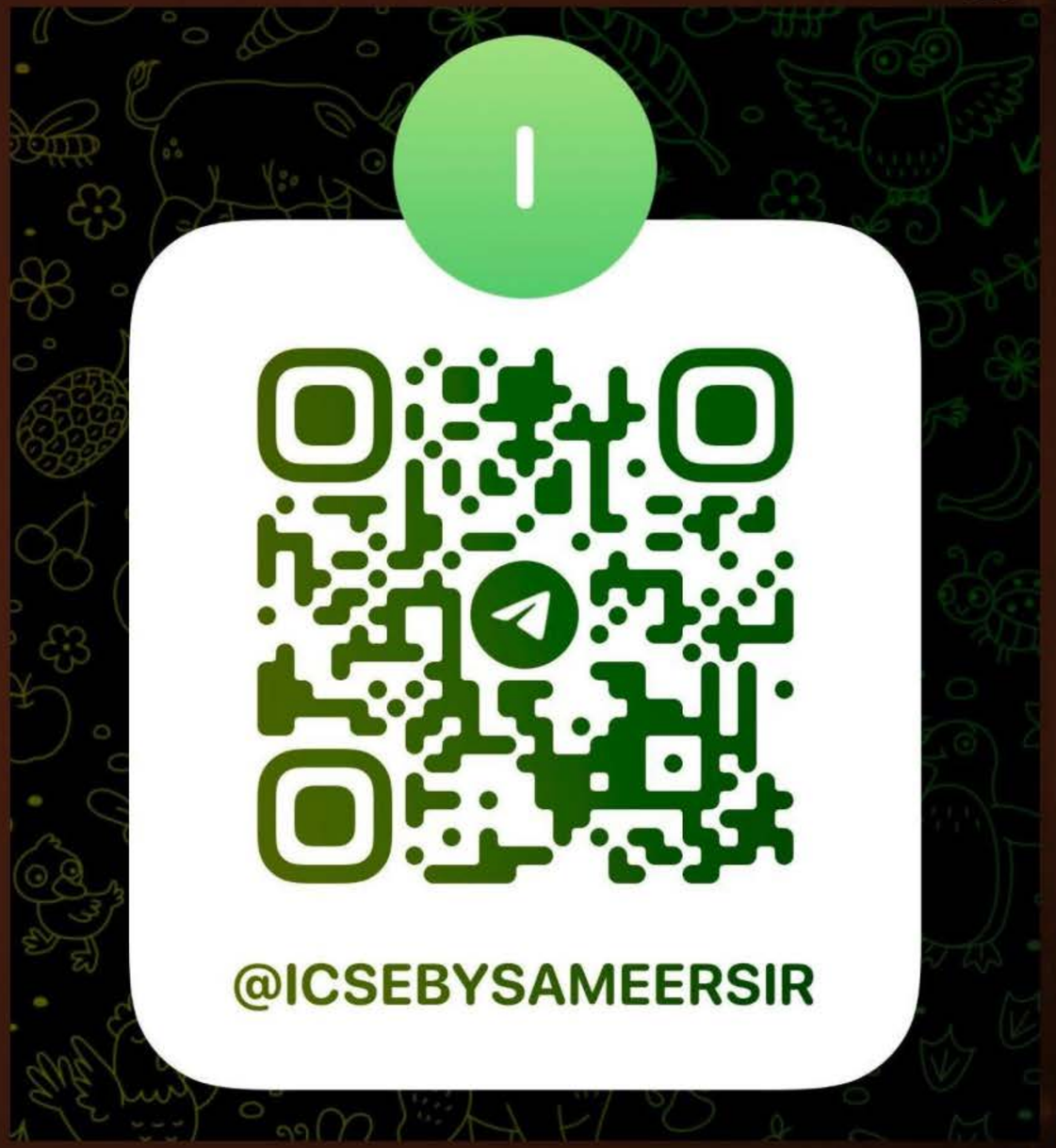
- *Iron ore, Manganese, Copper, Bauxite – uses and their distribution. (two leading states and their prominent mining areas)*
- *Conventional Sources: Coal, Petroleum, Natural gas (distribution, advantages and disadvantages).*
- *Hydel power - Bhakra Nangal Dam and Hirakud Projects and their benefits.*
- *Non-conventional Sources: Solar, wind, tidal, geo-thermal, nuclear and bio-gas (important areas and advantages and disadvantages).*

Fail → V stopped trying



Telegram

https://t.me/icsebysameersir



Fossil Fuel

Peat
Lignite
Bituminous
Anthracite

Gondwana
Tertiary



Coal

Assam - Arakan - Digboi
Gujarat - Cambay - Ankleshwar
Kalol
Mumbai High → Offshore



Crude Oil

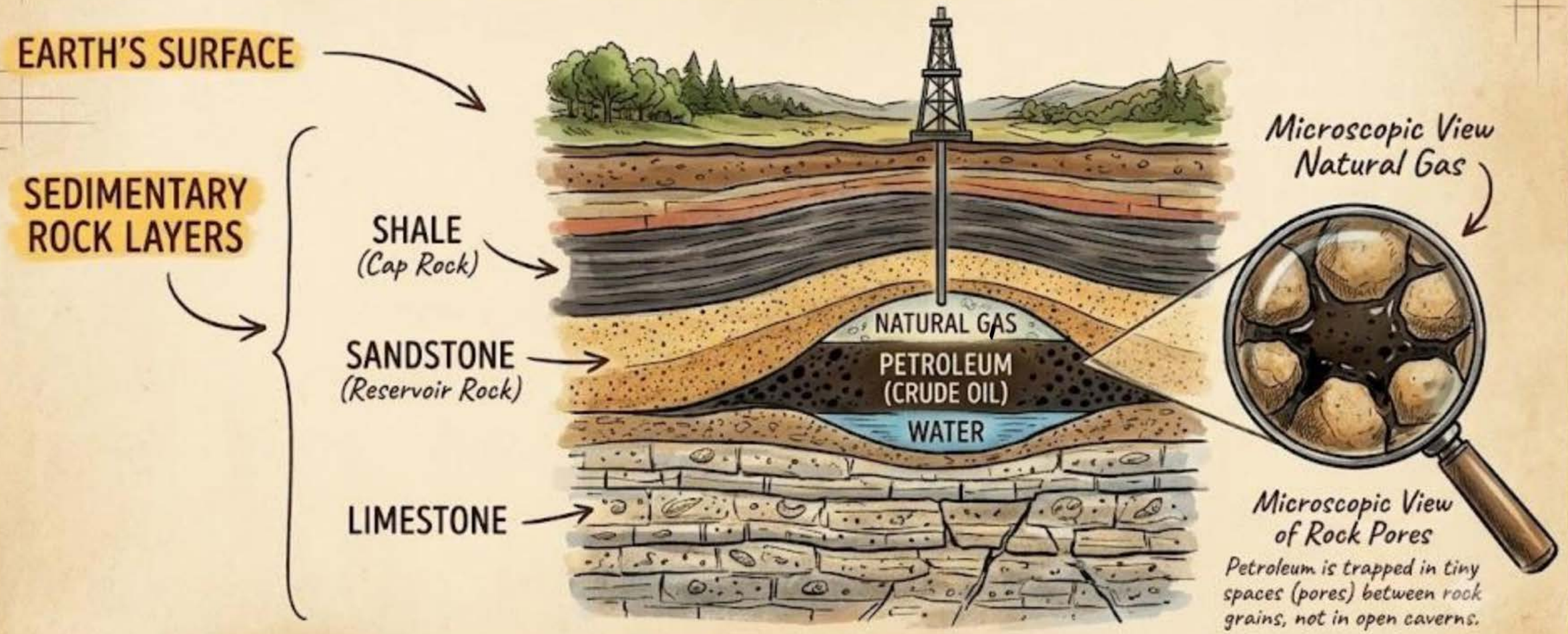
LPG
CNG



Natural Gas

WHERE PETROLEUM IS FOUND UNDERGROUND

Reservoirs in Sedimentary Rock Formations

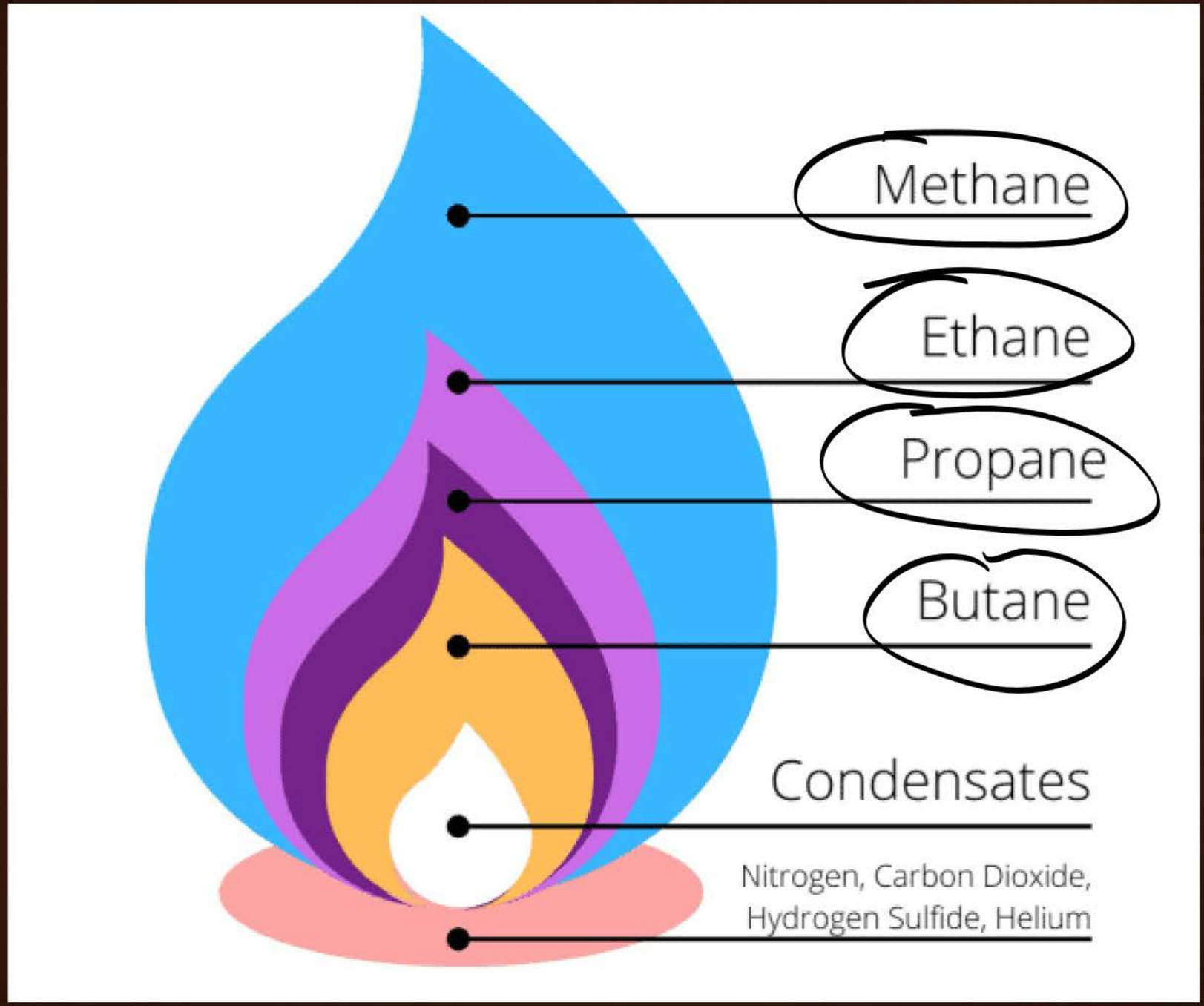


Petroleum is found in underground reservoirs in sedimentary rock formations like sandstone, shale and limestone.

Natural Gas

Inflammable } ज्वलनशील
Flammable }

- Natural gas occurs in association with mineral oil.
- Therefore, an oilfield yields natural gas almost invariably.
- **It is mainly composed of methane (95%) with small amounts of propane and ethane.** Butane
- It is a fossil fuel Natural gas deposits mostly accompany oil deposits because it has been formed by decomposing remains of dead animals and plants buried under the earth.


Methane
Ethane
Propane
Butane
Condensates
Nitrogen, Carbon Dioxide,
Hydrogen Sulfide, Helium

Natural Gas



Refining of Crude oil

- **Liquefied Petroleum Gas (LPG):** The gas supplied for household use is called LPG (Liquefied Petroleum Gas) and is a by-product obtained after refining crude oil.
- The main component of LPG is butane, the other being propane and ethane.
- It is odourless, but the LPG in our domestic gas cylinders gives a foul smell.

Natural Gas

- This is, in fact, due to ethyl mercaptan, a foul smelling gas, added to LPG so that any leakage of LPG from the cylinder can be detected instantaneously.
- In many places, LPG is being replaced by Piped Natural Gas (PNG), which is not stored in a cylinder like LPG but is continuously supplied through pipeline from the source.

Liquefied Petroleum Gas (LPG)



Natural Gas

Compressed Natural Gas (CNG): The gas used for running vehicles is known as Compressed Natural Gas (CNG) and as the name suggests it is obtained from natural gas.

- It is being used as an alternative to petrol and diesel for vehicles.

	LPG	CNG	PNG
Full form	Liquefied Petroleum Gas	Compressed Natural Gas	Piped Natural Gas
Source	By-product of crude oil refining (also from gas)	Natural gas, compressed at high pressure	Natural gas through pipelines
Main gas	Butane, propane	Methane	Methane
Use	Household cooking (cylinders)	Vehicle fuel (cars, buses, autos)	Household and industry via pipes
Note	Ethyl mercaptan added to give smell and detect leaks	Cleaner than petrol/diesel; Delhi runs public transport on CNG	No cylinder needed; continuous supply

Distribution

Over three-fourths of India's natural gas comes from

- **Mumbai High;**
- **the rest is produced in Assam, Tamil Nadu, Rajasthan and Tripura.**

Advantages

(i) Compared to petroleum or coal, natural gas is an eco-friendly fuel as it causes less damage to the environment. It is made up of methane and results in less carbon emissions.

(ii) It is easier to preserve than other fuels. It can be stored and transported through pipelines, small storage units, cylinders or tankers on land and sea.

(iii) It can be piped into houses for heating and cooking purposes and running a variety of appliances.

Advantages

(iv) It is a cleaner, cheaper fuel than diesel or gasoline. – petrol

(v) It is lighter than air and tends to dissipate when there is a leakage. Spread

(vi) It is used for producing hydrogen, ammonia for fertilizers and some paints and plastics.

Disadvantages

(i) Leaks of natural gas are difficult to detect and are very dangerous. Such leaks may cause explosions or fire. When inhaled, the gas is highly toxic. Leakage of natural gas can have serious consequences as methane is more toxic than carbon dioxide.

(ii) For use as fuel, except for methane, all other constituents of natural gas have to be extracted. Processing results in many by-products: hydrocarbons (propane, ethane etc.), sulphur, water, helium, nitrogen and carbon dioxide.

Disadvantages

(iii) Like all fossil fuels natural gas, though found in abundance, is non-renewable and hence likely to be exhausted at some point of time.

(iv) The infrastructure for natural gas production and distribution is fairly expensive. This includes separate plumbing systems and specialised tanks.

1. Which of the following is considered the cleanest burning fossil fuel?

A) Coal

B) Petroleum

C) Natural Gas

D) Lignite

2. Natural gas is predominantly composed of which gas?

A) Propane

B) Butane

C) Ethane

D) Methane

7. Which of the following is a major advantage of natural gas over coal?

A) It leaves no ash or solid residue behind

B) It is a renewable resource

C) It is difficult to transport via pipelines

D) It produces excessive smoke

8. In the transport sector, natural gas is compressed and used as an eco-friendly alternative to petrol and diesel. What is it called?

A) Clean Natural Gas (CNG)

B) Compressed Natural Gas (CNG)

C) Carbon Natural Gas (CNG)

D) Combustible Natural Gas (CNG)

9. Which of the following is a significant disadvantage of natural gas?

A) It emits large amounts of sulphur dioxide

B) It produces excessive solid waste

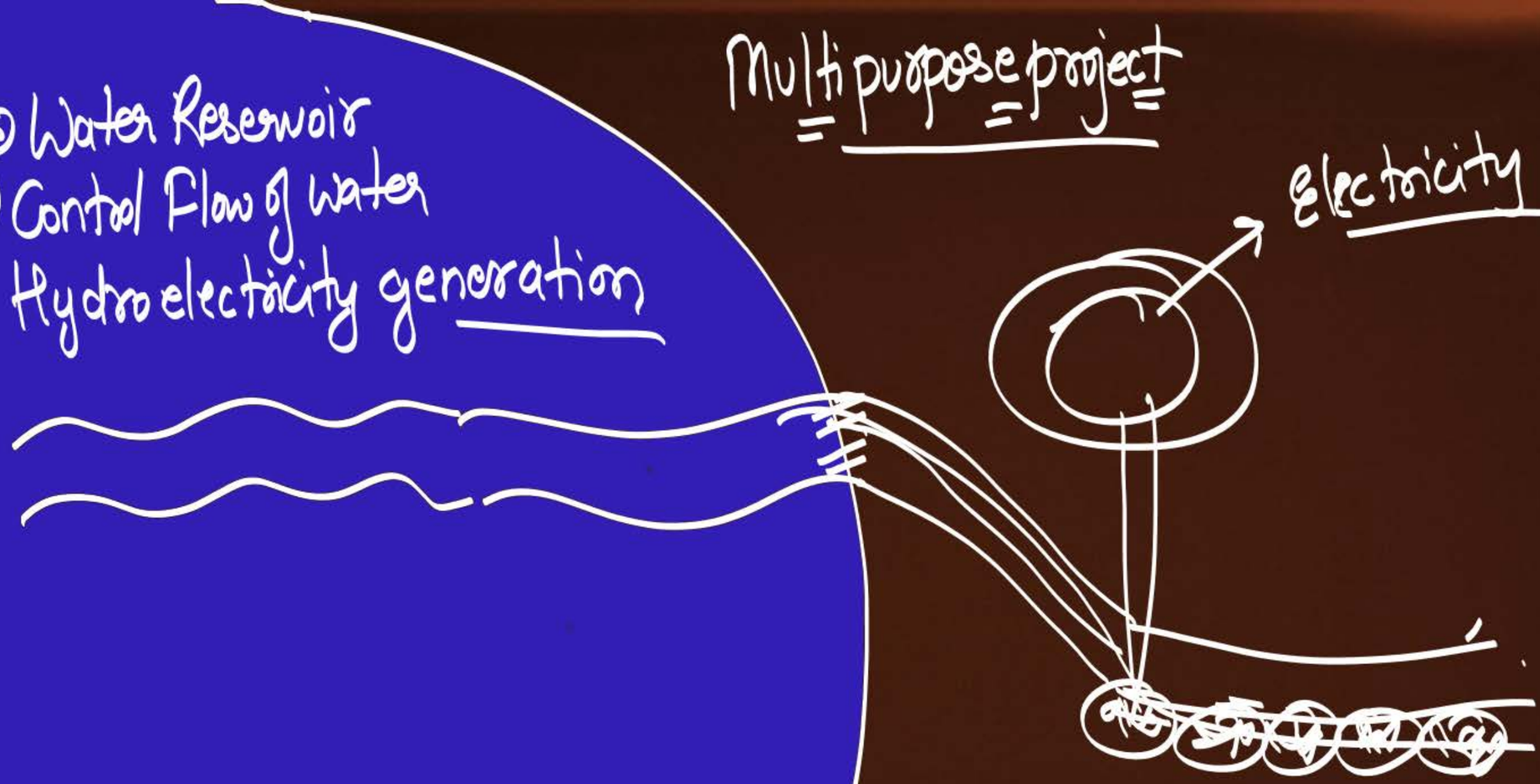
C) It is highly inflammable and prone to dangerous leakages

D) It has a very low calorific value


Hydroelectricity

Multipurpose project

- Water Reserwoir
- Control Flow of water
- Hydro electricity generation



Hydel Energy
Hydro Electricity

Hydel Power

- Electricity generated from water is called hydel power or hydroelectricity.
- Hydroelectricity is produced from the energy that is released when water falls from a high level with great force.

Hydel Power

- The water flowing in a river is collected by constructing a big dam where the water is stored.
- Then it is allowed to fall from a height. The blades of the turbine located at the bottom of the dam move with the force of falling water, which in turn rotates the generator and produces electricity.
- Hydro power does not cause any pollution, it is renewable and normally the hydro power projects are multi-purpose projects which are used to control floods, for irrigation, navigation, etc.

Hydel Power

- The total power generation capacity of the hydroelectric power plants depends on
- the head of water(Height of the falling water) and
- volume of water flowing towards the water turbine.

- It is the most widely used form of renewable energy. Once a hydroelectric complex is constructed, the project produces no direct waste, and has a considerably lower output level of the greenhouse gas, carbon dioxide (CO_2) than fossil fuel powered energy plants.

Hydel Power

- India was one of the pioneering countries in establishing hydro-electric power plants.
- The power plant at Darjeeling and Shimsha (Shivanasamudra) was established in 1898 and 1902 respectively.

West Bengal

Karnataka

Advantages

(i) Hydel power is a clean, non-polluting source of energy. The water used does not contaminate air or water by producing harmful wastes or poisonous bi-products.

(ii) It does not produce any Greenhouse Gases and protects us from the harmful effects of Global Warming.

(iii) It is a renewable source of energy which can be used again and again.

(iv) The dams built to produce hydroelectricity help to conserve and store water.

(v) It is economical and sustainable as it saves on fossil fuels and is cheaper than electricity produced from fossil fuels and nuclear power.

Disadvantages

⊙ Affects aquatic life.

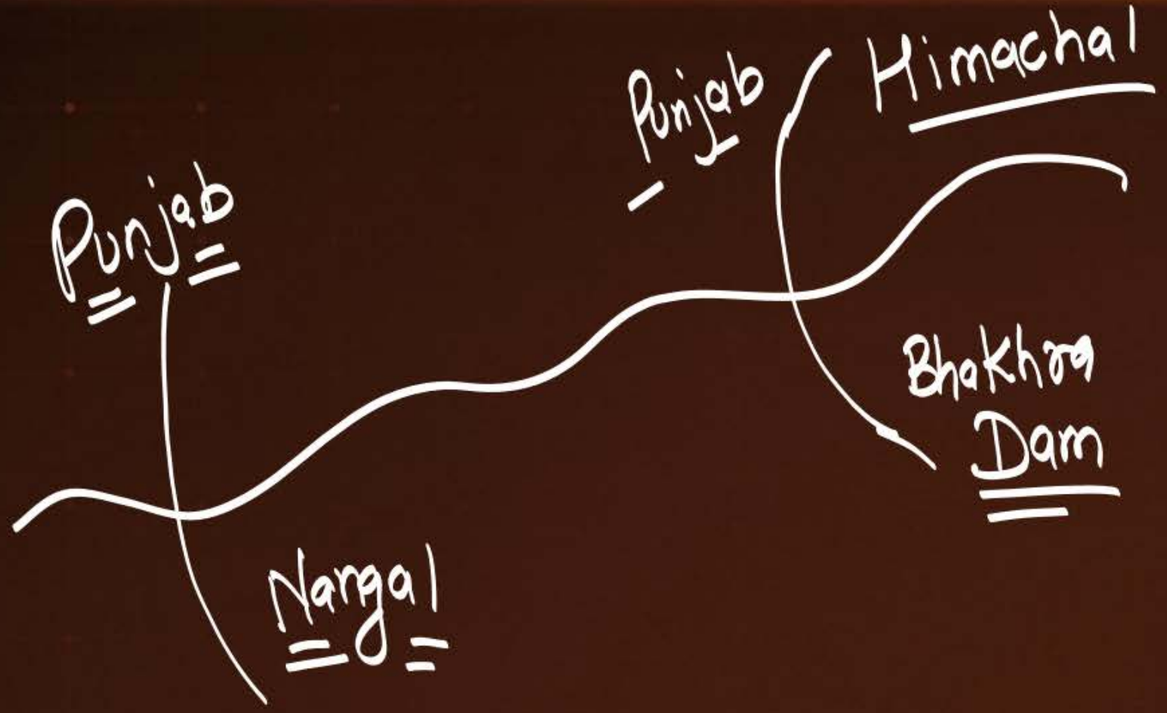
(i) The cost of building dams for producing hydroelectricity is quite high.

(ii) Building a large dam alters the natural water-table level and could change natural flow of river.

(iii) Building a large dam can cause serious geological damage like triggering earthquakes in the area.

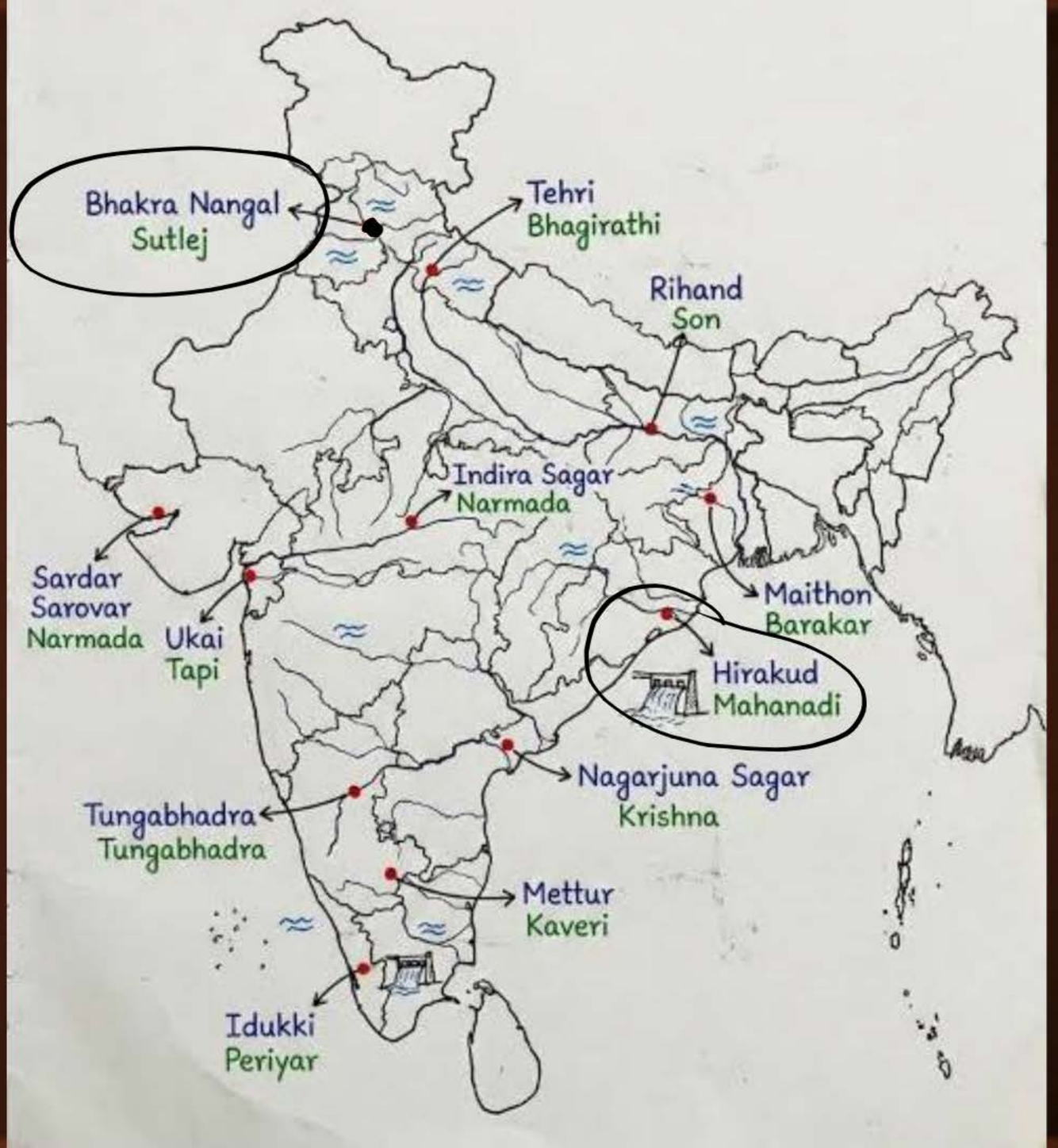
(iv) Building of dams to produce electricity, displaces many people from the area and disturbs their life physically, mentally and psychologically. Socio-economic

Project	River	State(s)	Key facts
Bhakra-Nangal	Sutlej	Punjab/Himachal Pradesh (benefits Punjab, Haryana, Rajasthan)	Reservoir = Gobind Sagar (about 90 km long); Nangal dam supplies Bhakra canal system; joint venture; irrigation, power, flood control
Hirakud	Mahanadi	Odisha (near Sambalpur)	One of the first major multipurpose projects after independence; controls Mahanadi delta floods; one of the world's longest dams


Punjab
Punjab
Himachal
Bhakhra
Dam
Nangal

Bhakra Nangal Dam

- The Bhakra Nangal Project is a joint venture of Punjab, Haryana and Rajasthan Governments.
- The two dams, one at Bhakra and another at Nangal, together is referred to as Bhakra-Nangal Project.
- The Bhakra-Nangal project comprises the following:
  - Two dams at Bhakra and Nangal
  - Nangal hydel plant
  - Power houses
  - Bhakra canal system.


Bhakra Nangal
Sutlej
Tehri
Bhagirathi
Rihand
Son
Indira Sagar
Narmada
Sardar
Sarovar
Narmada
Ukai
Tapi
Maithon
Barakar
Hirakud
Mahanadi
Nagarjuna Sagar
Krishna
Tungabhadra
Tungabhadra
Mettur
Kaveri
Idukki
Periyar


Gobind Sagar
Bhakhra



Bhakra Nangal Dam

Bhakra Nangal Dam

1. **The Bhakra Dam** is built across the Sutlej river near the border between Punjab and Himachal Pradesh. The 90km long reservoir of Bhakra dam is known as Gobind Sagar. It has a storage capacity of 9.3billion cubic meters.

2. **The Nangal Dam** has been constructed at Nangal on the river Sutlej in Punjab. It supplies water to Bhakra irrigation canals.

Bhakra Nangal Dam

3. **Power Houses** four power houses at Ganguwal, Kotla, left bank power house and the right bank power house have a combined installed capacity of 1204MW.

4. **Bhakra Canal System** provides irrigation facility to 10million acres of land in Punjab, Haryana and Rajasthan.

   The aim of these projects are:

   - to provide water for irrigation,
   - to generate hydro-electricity, and
   - to prevent flooding from Sutlej-Beas rivers.

Bhakra Nangal Dam

Irrigation, Water Storage, Electricity generation

Benifits

- The dam holds excess waters during the monsoon and provides a regulated release during the year.
- The dam provides irrigation to fields in Punjab, Haryana, and Rajasthan.
- The Bhakra main line is a canal that mostly supplies irrigation water to Haryana.
- Bhakra Dam has ten power generators with five on each side.

Bhakra Nangal Dam

The power generated at Bhakra Dam is distributed among partner states of Punjab, Haryana, Rajasthan, Himachal Pradesh, Chandigarh and Delhi.

- Three additional power plants are on the two canals Nangal Hydel Channel and Anandpul Sahib Hydel Channel that originate from Nangal dam.



Hirakud Dam

- Hirakud Dam is built across the Mahanadi River, about 15km from Sambalpur in Odisha.
- It is one of the first major multi-purpose river valley projects started after India's independence.
- The dam regulates the flow of the Mahanadi river and produces hydroelectricity through several hydroelectric plants.

Hirakud Dam

- The dam helps control floods in the Mahanadi delta and irrigates 75,000sq km of land.
- The project provides kharif and rabi irrigation in districts of Sambalpur, Bargarh, Bolangir, and Subarnpur.
- The water released by the power plant is used to irrigate areas in Mahanadi delta.



12. Which of the following is a major advantage of hydel power?

A) It is non-renewable

B) It is an inexhaustible and renewable source of energy

C) It causes high air pollution

D) It requires the continuous purchase of raw materials

13. On which river is the Bhakra Nangal Dam constructed?

A) Ganga

B) Sutlej

C) Mahanadi

D) Narmada

14. Which states are the primary beneficiaries of the power and irrigation from the Bhakra Nangal Project?

A) Gujarat, Maharashtra, and Madhya Pradesh

B) Odisha, West Bengal, and Bihar

C) Punjab, Haryana, and Rajasthan

D) Karnataka, Kerala, and Tamil Nadu

15. On which river is the Hirakud Dam built?

A) Godavari

B) Krishna

C) Mahanadi

D) Cauvery

20. Which of the following is considered a socio-economic drawback (disadvantage) of building large hydroelectric projects?

A) Flood control

B) Large scale displacement of local communities and tribes

C) Promotion of inland navigation

D) Development of pisciculture (fish farming)

P4Q

i) What are the benifits of hydel power? → (3)

ii) Why natural gas is more suitable than petroleum & coal? → (2)

Thank You